

WEATHERLAB

User Manual

"WEATHERLAB"

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USER MANUAL

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1. Welcome to WeatherLab

WeatherLab is a weather station based on a Raspberry Pi 5. It continuously measures the indoor climate and also shows the current weather and the forecast for its location. This manual explains how to operate the station, how to read the displays and how to use the Telegram bot. Technical details on the setup can be found in the Technical Documentation.

1.1 WHAT THE STATION CAN DO

- **Measure the indoor climate:** temperature, humidity, air pressure and air quality in the room, updated every second.
- **Show the outdoor weather:** current values and a 5-day forecast from OpenWeather, plus a weather map.
- **Record the history:** indoor and outdoor temperature for the last 7 days as a chart.
- **Give ventilation tips:** the station tells you whether opening the windows is worthwhile right now.
- **Traffic-light LEDs:** three LEDs show the air quality even from across the room.
- **Its own Wi-Fi:** phones and laptops can connect directly to the station.
- **Telegram bot:** sends alerts and a daily report to your phone every morning.
- **Settings:** Telegram recipients, Wi-Fi, locations and the PIN can be changed directly on the display, protected by a PIN.
- **Multiple locations:** the outdoor weather, forecast and maps can be switched to a different location.

1.2 THE COMPONENTS AT A GLANCE

COMPONENT	PURPOSE
Touch Display (7 inch)	shows the dashboard and is operated by finger
BME680 sensor	measures temperature, humidity, air pressure and air quality in the room
Three LEDs (green, yellow, red)	show the air quality like a traffic light
Raspberry Pi 5	the small computer that controls everything
LAN cable	connects the station to the internet (for outdoor weather and Telegram)
Power supply (USB-C)	power

2. Switching On and Off

2.1 SWITCHING ON

1. Plug the LAN cable into the Raspberry Pi so that the station has internet access.
2. Plug in the power supply. The Raspberry Pi starts automatically, there is no power button.
3. After about a minute, the dashboard appears automatically in full screen on the display.

Immediately after startup, some fields still show `--`. The indoor values appear after a few seconds, the outdoor values within a minute. The history and the pressure trend fill up over time.

2.2 SWITCHING OFF

To switch the station off, the power supply can simply be unplugged while it is running. When it is plugged in again, the station starts up normally and shows the dashboard after about a minute.

3. Connecting a Phone or Laptop

The station provides its own Wi-Fi network. Anyone connected to it can open the dashboard in a browser, just like on the display. This Wi-Fi also gives access to the internet, as long as the station is connected via the LAN cable.

3.1 CONNECTING TO THE WI-FI

1. Open the Wi-Fi settings on your phone or laptop.
2. Select the network **WeatherLab**. If the name has been changed in the settings (section 9.4), the new name is shown instead.
3. Enter the Wi-Fi password. It is deliberately not printed in this manual and can be obtained from the project team.



3.2 OPENING THE DASHBOARD IN A BROWSER

After connecting, open a browser such as Chrome, Safari or Firefox and enter one of these addresses:

ADDRESS	NOTE
http://weatherlab.lan/	easy to remember
http://192.168.10.1/	always works, even if the name is not resolved

Make sure to type `http://` at the beginning. Otherwise some browsers try `https://` and the page does not load. On a phone, it is worth saving the page as a bookmark or adding it to the home screen.

4. Using the Touch Display

4.1 THE NAVIGATION BAR

The navigation bar runs along the bottom edge of the display. Tapping an entry switches to that page. The time and date are shown on the far right, with the gear icon for the settings just to the left.

BUTTON	PAGE
Innen / Außen	home page with the indoor climate on the left and the outdoor weather on the right (section 5.1)
Aktuell	detailed outdoor weather with weather map (section 5.2)
Vorschau	5-day forecast and location map (section 5.3)
Verlauf	chart of the last few hours or days (section 5.4)
Pi-Stats	status of the Raspberry Pi (section 5.5)
⚙️ (gear icon)	settings, protected by a PIN (chapter 9). An orange dot on the gear icon means that someone is waiting to be approved for the Telegram bot.

4.2 NIGHT MODE AND AUTOMATIC RETURN

- **Night mode:** between 22:00 and 6:00 the display is dimmed. A tap wakes it up for 2 minutes.
- **Back to the home page:** if the display is not touched for 90 seconds, it automatically returns to the **Innen / Außen** page.

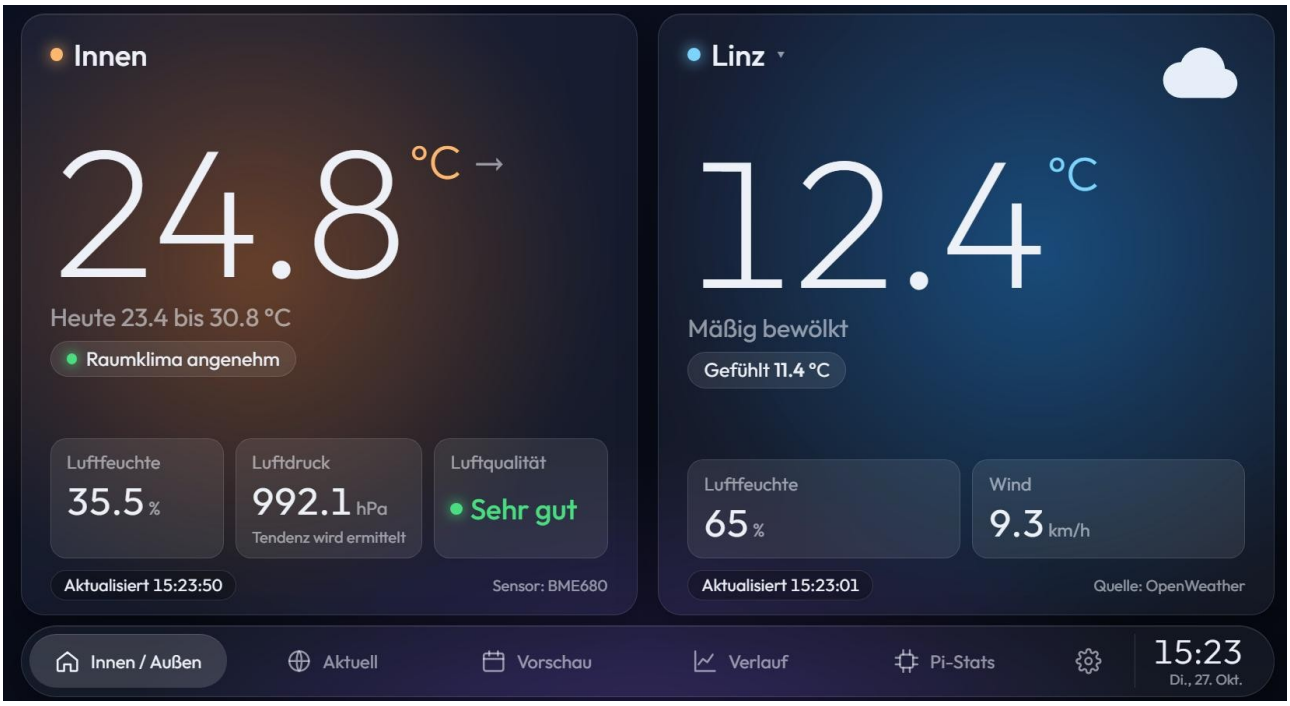
The project team can change both times if required.

5. The Dashboard Pages

5.1 INNEN / AUßEN (INDOOR / OUTDOOR)

The home page is split in two: the indoor climate from the sensor on the left, the outdoor weather from OpenWeather on the right.

The name of the selected location is shown at the top right, with a small arrow ▾. Tapping it opens the list of saved locations. Tapping a location switches to it immediately, no PIN is required. The outdoor weather, forecast and maps then show the new location. New locations can only be added in the settings (section 9.5).



DISPLAY	MEANING
Large temperature (left)	room temperature in °C
Arrow next to the temperature	trend compared with 30 minutes ago: ↗ warmer, ↘ cooler, → unchanged
Line below the temperature	lowest and highest room temperature of the day
Note with coloured dot	ventilation recommendation (section 6.3)
Luftfeuchte (humidity)	relative humidity in the room in %
Luftdruck (air pressure)	air pressure in hPa, with the trend over the last 3 hours below it (section 6.2)
Luftqualität (air quality)	level from “Sehr gut” (very good) to “Sehr schlecht” (very poor) (section 6.1)
Location name with ▾ (top right)	selected location, a tap opens the location picker
Large temperature (right)	outdoor temperature with the weather description below it

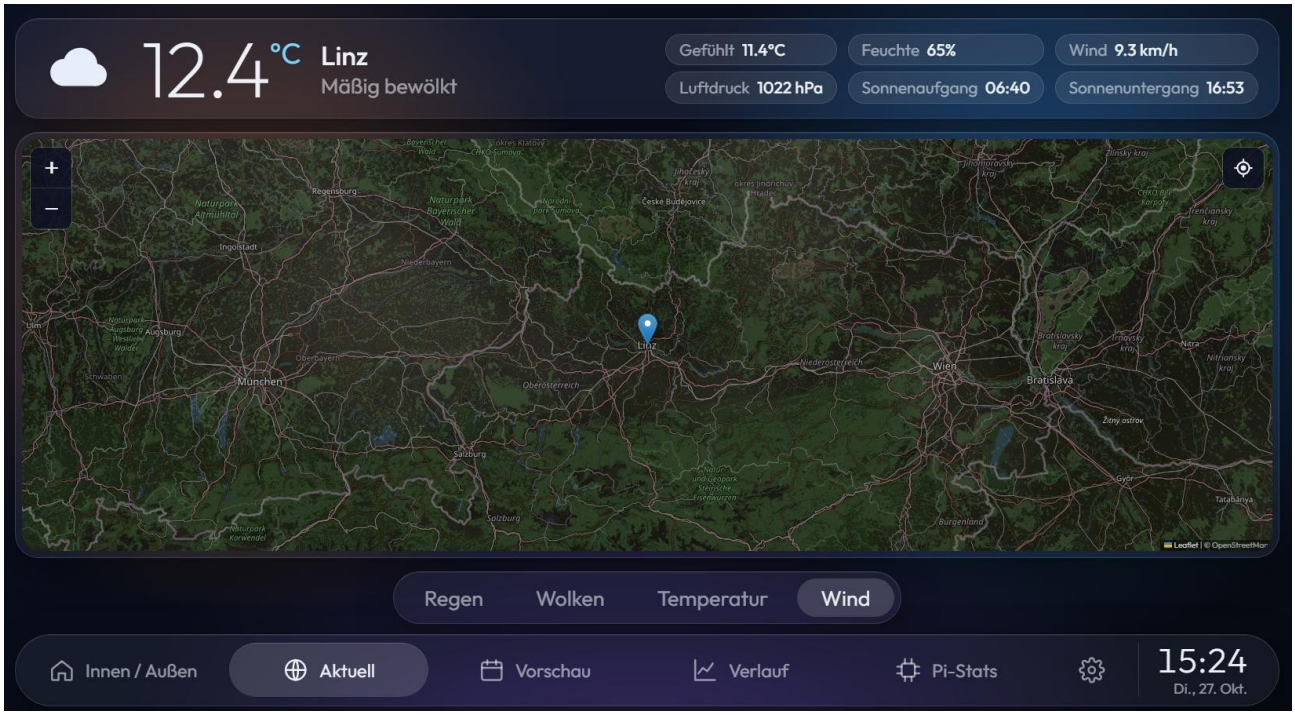
DISPLAY	MEANING
Gefühlt (feels like)	perceived outdoor temperature, for example colder in windy weather
Humidity and wind (right)	outdoor humidity in % and wind speed in km/h
Time stamp boxes at the bottom	time of the last measurement or query

5.2 AKTUELL (CURRENT)

The top section shows the outdoor weather with icon, temperature and description. Next to it, small boxes show the perceived temperature, humidity, wind, air pressure, sunrise and sunset.

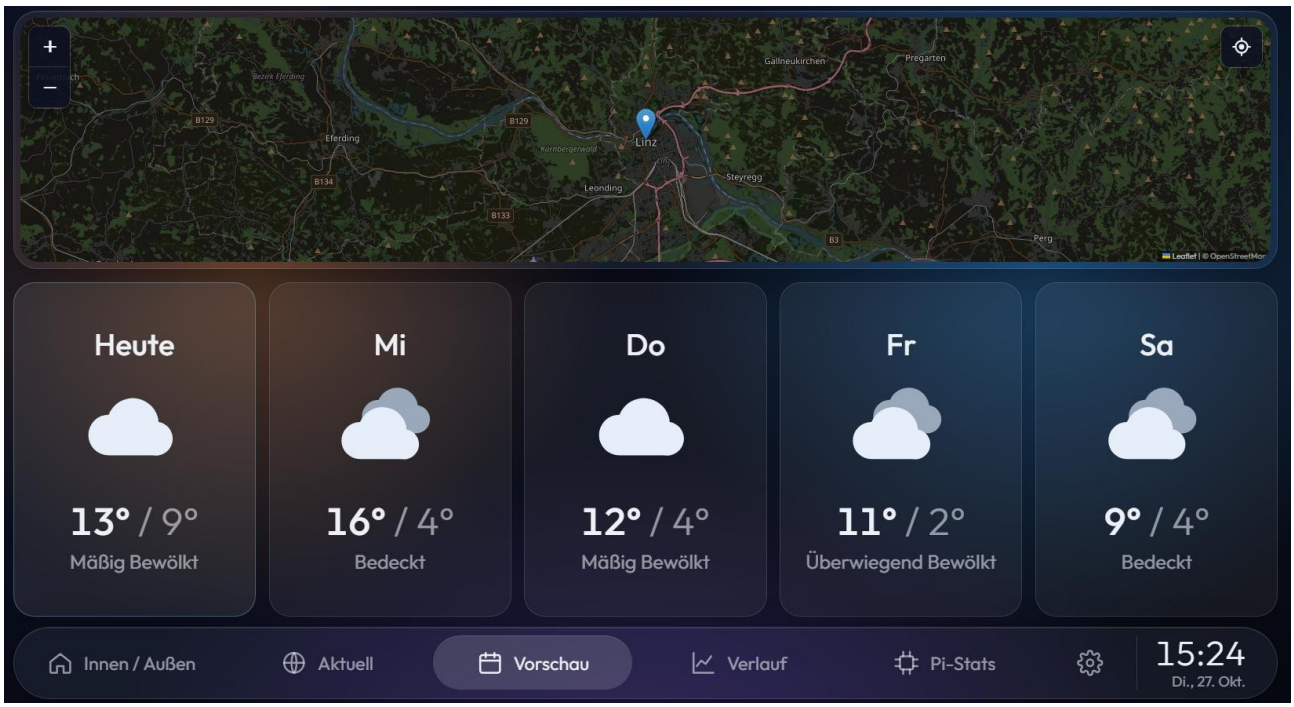
Below that is a live weather map. The buttons **Regen** (rain), **Wolken** (clouds), **Temperatur** and **Wind** switch between map layers. The **+** and **–** buttons at the top left zoom the map in and out, and it can be moved with one finger. The crosshair button at the top right brings the map back to the location.

The display only recognises one finger at a time, so pinch-to-zoom does not work. That is what the buttons are for. On a phone, both methods work.



5.3 VORSCHAU (FORECAST)

The map at the top shows the selected location, with the same + and – buttons and crosshair as on the Aktuell page. Below it are five cards for the coming days. Each shows the day of the week, weather icon, maximum and minimum temperature and a short description. Today is highlighted.



5.4 VERLAUF (HISTORY)

The chart shows indoor temperature, outdoor temperature and indoor humidity over time. Use the buttons at the top right to choose the period: **6 Std.** (6 h), **24 Std.** (24 h) or **7 Tage** (7 days). The boxes below the chart show the indoor and outdoor minimum and maximum as well as the average humidity.

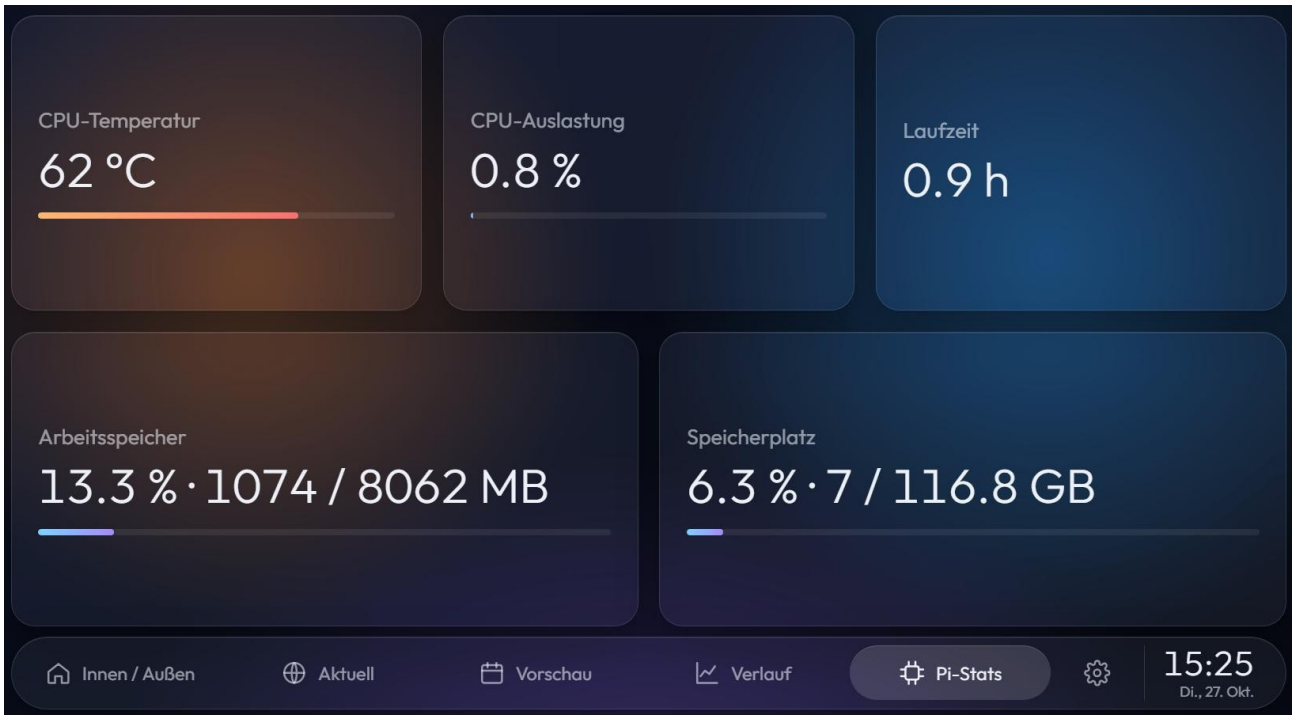
The station stores one data point per minute and keeps the values for 7 days. When it is first switched on, the chart is therefore still empty and fills up over time.



5.5 PI-STATS

This page shows how the Raspberry Pi itself is doing. It is not important for normal use but helps with troubleshooting.

DISPLAY	MEANING
CPU temperature	temperature of the processor, around 40 to 70 °C is normal
CPU load	how busy the processor is right now, in %
Uptime	time since it was last switched on
Memory	used and total RAM
Storage	used and total space on the memory card



6. Understanding the Readings

6.1 AIR QUALITY

The sensor measures how many gases are in the air, for example from stale breath, cooking fumes or emissions from materials. From this, the station calculates five levels:

LEVEL	MEANING	LED
Sehr gut (very good)	fresh, clean air	green
Gut (good)	pleasant indoor air	green
Mittel (moderate)	air is getting stale, ventilate soon	yellow / orange
Schlecht (poor)	stale air, ventilate now	red
Sehr schlecht (very poor)	very stale air, ventilate immediately	red

After it is first switched on, the sensor needs about 24 to 48 hours before the air quality readings are reliable. During this time, the levels may fluctuate.

6.2 AIR PRESSURE AND TREND

Below the air pressure, the display shows how it has changed over the last 3 hours. This gives a rough idea of the weather over the next few hours:

TREND	CHANGE IN 3 HOURS	WHAT IT USUALLY MEANS
rising sharply	more than +3 hPa	weather improving quickly
rising	+1 to +3 hPa	weather improving or staying fine
steady	−1 to +1 hPa	no major change
falling	−1 to −3 hPa	weather getting worse
falling sharply	more than −3 hPa	change in the weather, possibly storms or thunderstorms

The trend only appears once the station has been measuring for at least 3 hours.

6.3 VENTILATION RECOMMENDATION

The note with the coloured dot on the home page compares the indoor climate, air quality and outdoor weather. The station uses the absolute humidity for this, i.e. how much water is actually in the air. This is why ventilating can still make sense when it is raining, and achieve little when the sun is shining.

NOTE	WHAT TO DO
Raumklima angenehm (comfortable indoor climate)	nothing, everything is fine

NOTE	WHAT TO DO
Bitte lüften, die Luftqualität ist schlecht (please ventilate, air quality is poor)	open the windows
Lüften empfohlen, draußen ist die Luft trockener (ventilation recommended, the air outside is drier)	open the windows wide for a few minutes, this lowers the humidity
Luft feucht, Lüften bringt gerade wenig (humid air, ventilating won't help much right now)	it is similarly humid outside, try again later
Luft sehr trocken (air very dry)	brief ventilation is enough, plants or a humidifier may help
Raum sehr warm (room very warm)	above 26 °C, ventilate or provide shade
Raum kühl (room cool)	below 18 °C

6.4 HOW OFTEN THE VALUES ARE UPDATED

VALUE	UPDATE INTERVAL
Indoor values (sensor)	every second
Outdoor weather	every minute
5-day forecast	every 30 minutes
Verlauf	one data point per minute
Pi-Stats	every 5 seconds

7. Status LEDs

The three LEDs on the station show the air quality like a traffic light, so you can tell from across the room whether it is time to open the windows. Exactly one LED is always lit.

LED	AIR QUALITY	MEANING
green	Sehr gut or Gut	everything is fine
yellow / orange	Mittel (moderate)	ventilate soon
red	Schlecht or Sehr schlecht	ventilate now

If no LED is lit after switching on, this is normal. The sensor has to warm up first, which usually takes a few minutes.

8. Telegram Bot

The Telegram bot sends alerts and a daily report to your phone and answers questions about the current weather. All you need is the free Telegram app.

8.1 GETTING APPROVED

To prevent strangers from accessing the readings, the bot only responds to approved users.

1. Search Telegram for the station's bot (WeatherLabNotificationBot).
2. Tap **Starten** (Start) or send `/start`.
3. The bot replies that your request has been received.
4. Now someone with the PIN approves you on the display: an orange dot appears on the gear icon, and the request is listed under **Einstellungen → Telegram** (Settings → Telegram). A tap on **Freischalten** (Approve) is all it takes (section 9.3).
5. The bot sends "Du bist jetzt freigeschaltet" (you are now approved). From then on, you will receive alerts and the daily report.

If you cannot get to the station right now, you can also send your chat ID to the project team. The bot includes it in its reply to /start.



8.2 COMMANDS

All commands are also available via the menu icon next to the input field.

COMMAND	REPLY
/wetter	indoor and outdoor values at a glance, with a ventilation tip
/innen	indoor climate with trend, daily minimum and maximum and pressure trend

COMMAND	REPLY
/aussen	current outdoor weather with wind, sunrise and sunset
/vorschau	forecast for the coming days
/verlauf	minimum and maximum of the last 24 hours
/alarme	which alerts are currently active, plus the thresholds
/id	your own chat ID
/hilfe	list of all commands



8.3 ALERTS AND DAILY REPORT

The bot gets in touch automatically when a value crosses a threshold. Once everything is back to normal, it sends an all-clear. Each alert is sent only once, not every minute.

ALERT	WHEN
Innen zu warm (too warm indoors)	from 27 °C
Innen zu kalt (too cold indoors)	from 17 °C
Luftfeuchte zu hoch (humidity too high)	from 65 %, with a note on whether ventilating helps
Luft zu trocken (air too dry)	from 30 %
Schlechte Luft (poor air)	when the air quality is “Schlecht” or “Sehr schlecht”
Luftdruck fällt stark (pressure falling sharply)	by 3 hPa or more within 3 hours
Frost	from 0 °C outside
Starker Wind (strong wind)	from 50 km/h
Gewitter (thunderstorm)	when OpenWeather reports a thunderstorm
Regen (rain)	when it starts to rain outside

ALERT	WHEN
Schnee (snow)	when it starts to snow outside
Wetterstation meldet sich nicht (station not responding)	when no new readings arrive for 3 minutes

- **Rain, snow and thunderstorms:** the all-clear is only sent once it has been calm for 30 minutes. This avoids constant messages during short breaks.
- **Daily report:** every morning at 7:00 with the indoor climate, outdoor weather, ventilation tip and forecast.
- **Quiet hours:** between 22:00 and 7:00, alerts arrive without sound. They can still be seen in the chat in the morning.

9. Settings

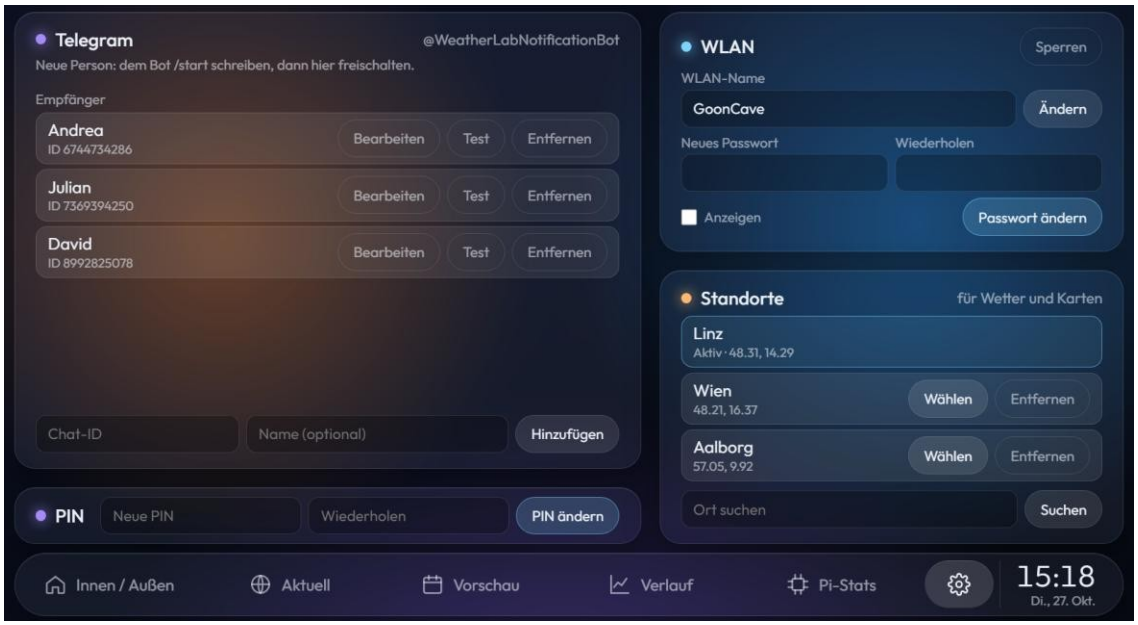
The settings (**Einstellungen**) are behind the gear icon at the bottom right of the navigation bar. This is where Telegram recipients, the Wi-Fi, locations and the PIN can be changed. To prevent unauthorised changes, they are protected by a PIN. The PIN is available from the project team and is deliberately not included in this manual.

9.1 OPENING AND LOCKING THE SETTINGS

1. Tap the gear icon ⚙️.
2. Enter the PIN using the number keys and tap **OK**.

Good to know:

- **Locking:** the **Sperren** (Lock) button at the top right locks the settings again. They also lock automatically as soon as another page is opened, which includes when the display has not been touched for 90 seconds.
- **Wrong PIN:** after 5 wrong attempts, input is blocked for 2 minutes.
- **Orange dot on the gear icon:** someone has sent /start to the Telegram bot and is waiting to be approved.



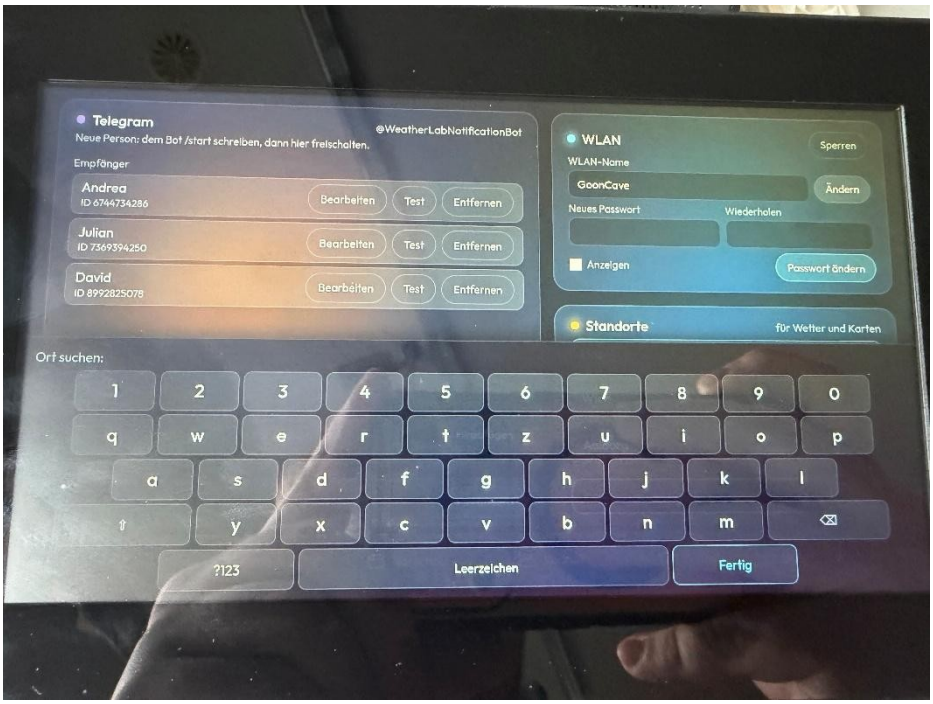
9.2 ON-SCREEN KEYBOARD

When you tap an input field on the display, a keyboard appears at the bottom. Above the keys you can see which field is being edited and what has already been typed. Passwords are shown as dots only.

KEY	EFFECT
⇧	next letter in upper case
?123 / ABC	switches between letters and special characters

KEY	EFFECT
⌫	deletes the last character
Leerzeichen	inserts a space
Fertig	closes the keyboard (done); in the location search it also starts the search

For chat IDs and the PIN, only a numeric keypad is shown. On a phone or laptop, the device's normal keyboard appears instead.



9.3 TELEGRAM RECIPIENTS

All people who receive messages from the bot are listed on the left. New requests appear above them, highlighted in orange.

BUTTON	EFFECT
Freischalten	accepts a new request, the person receives messages from now on
Ablehnen	rejects and deletes the request
Bearbeiten	changes the name: the ID and name appear in the form below. Type the new name, then tap **Speichern** (Save). **Abbrechen** (Cancel) leaves everything as it was.
Test	sends a test message to the person
Entfernen	tap twice ("Sicher?" – sure?), the person then no longer receives messages

Adding manually: enter the chat ID and, if desired, a name at the bottom and tap **Hinzufügen** (Add). Groups have a chat ID that starts with a minus sign.

9.4 WI-FI NAME AND PASSWORD

The Wi-Fi section shows the current network name of the station.

- **Wi-Fi name:** type the new name and tap **Ändern** (Change). Between 1 and 32 characters are allowed, without umlauts.
- **Password:** enter the new password under **Neues Passwort** (new password) and **Wiederholen** (repeat) and tap **Passwort ändern** (change password). It needs 8 to 63 characters; umlauts such as ä or ß are not allowed. With **Anzeigen** (show) you can see what you have typed.

After a change, the Wi-Fi restarts. All phones and laptops are briefly disconnected and have to reconnect using the new name or password. The display keeps running because it is not connected via Wi-Fi. Please let the project team know about a new password.

9.5 LOCATIONS

This is where you choose the location used for the outdoor weather, forecast and maps. The active location has a blue frame.

1. Type a place name into the **Ort suchen** (search location) field, for example Wien.
2. Tap **Fertig** or **Suchen** (Search). A list of matching places appears.
3. Tap **Wählen** (Select) next to the right place. It is saved and becomes active immediately.

Saved locations have two buttons:

- **Wählen:** makes a saved location the active one.
- **Entfernen:** tap twice. At least one location must always remain.

You can also switch between saved locations without a PIN directly on the home page (section 5.1). After switching, it takes a few seconds until the outdoor weather and forecast for the new location are available. The indoor values do not change, as they always come from the sensor in the room. From the moment of the switch, the history records the outdoor temperature of the new location.



9.6 CHANGING THE PIN

The small **PIN** box below Telegram changes the PIN for the settings.

1. Under **Neue PIN**, enter a new PIN with 4 to 8 digits.
2. Under **Wiederholen**, enter the same PIN again.
3. Tap **PIN ändern** (change PIN).

The new PIN takes effect immediately, on the display and on phones. Please inform the project team, otherwise nobody will be able to open the settings.

10. Troubleshooting

10.1 COMMON PROBLEMS

PROBLEM	POSSIBLE CAUSE	SOLUTION
Display is dark	night mode between 22:00 and 6:00	tap the display, it stays bright for 2 minutes
Fields show - -	station is starting up or the sensor is still being read	wait a minute
Outdoor values missing	no internet, for example LAN cable unplugged	check the LAN cable
History is empty	station has not been running for long	wait, it fills up every minute
Air pressure without trend	there are not yet 3 hours of readings	wait
Air quality jumps a lot	sensor still in its burn-in period (24 to 48 hours)	wait
WeatherLab Wi-Fi not visible	station is off or still starting up	wait a minute, then search again
Page does not load in the browser	wrong address, or https instead of http	enter http://192.168.10.1/
Display only shows the desktop	the full-screen view was closed	restart the station
Bot replies “nicht freigeschaltet” (not approved)	the request has not been approved yet	send /start and have it approved in the settings (section 8.1)
Bot does not reply at all	station or internet is down	check the station and the LAN cable
Map cannot be zoomed with two fingers	the display only recognises one finger	use the + and − buttons on the map
Outdoor weather shows a different place	a different location is selected	tap the location name on the home page and select the right location
No Wi-Fi after a change	the Wi-Fi name or password was changed in the settings	connect using the new name or password
Settings will not open	wrong PIN or 5 failed attempts	wait 2 minutes, ask the project team for the PIN
New features missing on a phone or PC	the browser is still showing the old version of the page	reload the page, press Ctrl+F5 on a PC

If none of this helps, please inform the project team and describe what is shown on the display.

10.2 CARE AND NOTES

- Do not place the station directly next to a radiator or window or in direct sunlight, otherwise the sensor will give false readings.
- Do not cover the sensor or breathe on it.
- Clean the display only with a dry, soft cloth.
- Do not share the Wi-Fi password or write it down publicly.
- Do not share the PIN for the settings either.